How the chaos of the Three-Body Problem determins the shape of Saturn rings

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6th March 2025

Word count:1628

1 Introduction

The three-body problem is an astronomical unsolvable case of three massive bodies orbiting one another. Being the basis for 300 years of studies, multiple popular science books, and inspiration for movies and shows, it is the most popular case of a dynamical system of chaotic behaviour (Musialek, Quarles, 2014). The importance of understanding the three-body problem is the fact that it naturally occurs in celestial mechanics, such as the rings of Saturn. Their complex structures, including gaps, waves, and sharp edges, are the result of the interaction between the ring belt, Saturn, and its "shepherd" moons. In this essay, I will strive to depict and analyse the causes of the gaps in Saturn's rings due to the chaos of the Three-Body Problem from the interaction of ring particles with the moon Mimas. The main two parts of the rings that will be analysed are the most stable A ring and Cassini Division, which is a large gap between ring A and B.

2 Three body problem - mathematical formulation

To formulate the Three-Body Problem in celestial mechanics, all we need are Newton's gravitational equations of motion:

$$F = \frac{d^2x}{dt^2}m = G\frac{mM}{r^2} \rightarrow \frac{d^2x}{dt^2} = G\frac{M}{r^2} \tag{1}$$

where G is the gravitational constant; m, and M masses of two objects and r is the distance between them. In the inertial coordinate system, suppose that we consider three bodies A, B and C and their coordinates q_a, q_b and q_c. The distance between A and B will therefore be q_{ab}, between AC - q_{ac} and so on for

each pair. Rewriting the Newton equation for each of the bodies and considering forces to be vector quantities, we get the following.

$$\begin{aligned}
\frac{d^2q_A}{dt^2} &= GM_B\frac{(q_B-q_A)}{r_{AB}^3} + GM_C\frac{(q_C-q_A)}{r_{AC}^3} \\
\frac{d^2q_B}{dt^2} &= GM_A\frac{(q_A-q_B)}{r_{AB}^3} + GM_C\frac{(q_C-q_B)}{r_{BC}^3} \\
\frac{d^2q_C}{dt^2} &= GM_A\frac{(q_A-q_C)}{r_{AC}^3} + GM_B\frac{(q_B-q_C)}{r_{BC}^3}
\end{aligned} \tag{2}$$

where q is the coordinates of each body in the inertial coordinate system and so $q_n - q_m$ are the relative positions of the masses(Barrow-Green). Contrary to the situation for two bodies, this system of equations is analytically unsolvable. There is no general solution; chaos appears, which we will later use in the context of Saturn's rings. As there is no general solution, the situation will be reduced to specific numerical computations. These special cases' trajectories and their stability will be used to show how even though the system is chaotic, there are still stable rings that orbit Saturn

To analyse the behaviour of the rings, we will also need a Poincaré section. It is a map of points that are taken from the trajectories of motion crossing an arbitrarily chosen plane. At last, we will also need the Lyapunov exponent, which measures the time rate at which an infinitesimally small distance between two points grows. Since all the points at Saturn's rings are affected by the three-body problem, the whole ring system is chaotic. Yet, we can clearly see and distinguish the rings. The answer to why it is so is the stability. Using a Lyapunov exponent at different distances from Saturn will help us understand why in some places there are rings while in others there are empty spaces. The exponent follows:

$$F^t(x_0+\varepsilon) - F^t(x_0) \approx \varepsilon e^{\lambda t} \tag{3}$$

Where the exponent is the λ and the left-hand side is the distance between two points after t steps. If $\lambda > 0$, the system grows indefinitely, and if $\lambda < 0$, it does not - the system settles down to a periodic trajectory. Rearranging the equation for the exponent and setting it for limits as $t \to \infty, \varepsilon \to 0$ we get:

$$\lambda = \lim_{t\to\infty} \frac{1}{t}\sum_{i=0}^{t-1} \ln|\frac{dF}{dx}|_{x=x_i}| \tag{4}$$

The exponent is thus an average time of $\ln|\frac{dF}{dx}|$ at a chosen state in the system (mathlib).

3 Saturn rings analysis

At first, we can compare the Poincaré sections of the most stable and the least stable point. The distances from Saturn will then be: for stable ring A r=1.32e8, and unstable Cassini division r=1.2e8. Simulating the Poincaré section of the Three Body Problem in Python for the values of trajectory crossing $\theta = 0$, we get the following graphs:

Figure 1: r=1.32e8

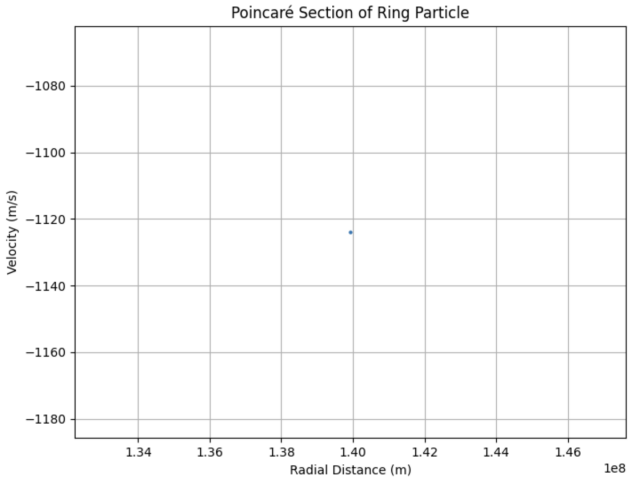


Figure 2: r=1.2e8

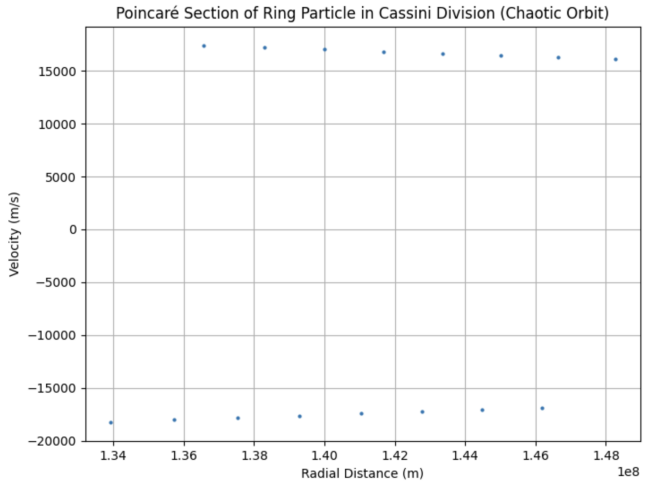


Figure 1 shows a single point. It means that each time the particle crosses a

plane $\theta = 0$, it has the same velocity at the same position. It is a perfect example of a stable orbit. In comparison, Figure 2 is a set of points that form two lines of positive and negative speeds at different distances from Saturn. The two different signs of velocity mean that each time the particle crosses $\theta = 0$, it either goes in one direction or the other. It does not follow a stable orbit like Figure 1, however, it is not yet an expected chaotic behaviour. The answer to why it is not chaotic is the limited time of computation. The simulation for both Poincare sections considered the trajectories at a time of 300 days with a time step of 100 seconds. And it took a few minutes to produce the graph. Increasing the simulation time or decreasing the time step exponentially increased the time it took to make the graph.

The next step in our analysis is to consider the divergence of two close trajectories over time in the same cases of r and then get the Lyapunov exponent over the distance from Saturn.

Figure 3: r=1.32e8

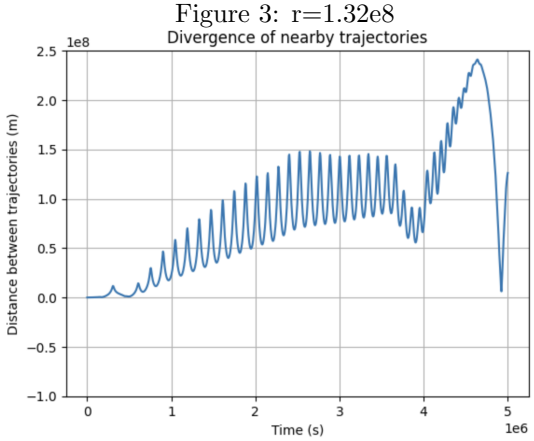
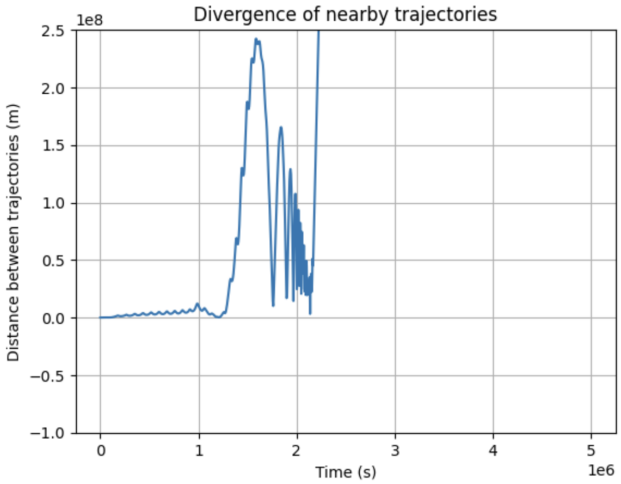


Figure 3 shows that for about 4 million seconds, the trajectories are close to each other but slowly diverging. After this time, they start to oscillate a little, and finally, they completely explode after the 5 millionth-second mark. It shows that the orbit even inside the A ring is chaotic. Now, let's compare it with the simulation for the Cassini division (Figure 4).

Compared to the A ring, slightly displaced trajectories in the distance from Saturn corresponding to the Cassini division almost immediately go into resonance. They oscillate around the point 0 until 2.1 million seconds, then fully diverge 3 million seconds earlier than the trajectories in the A ring.

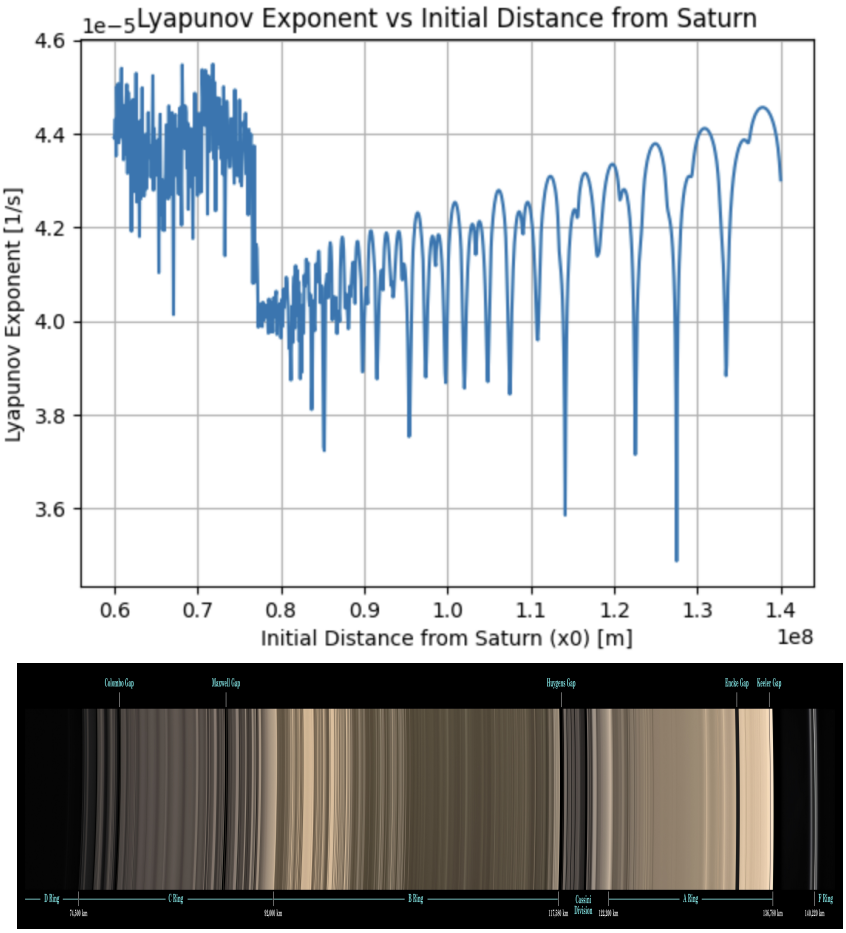
Figure 4: r=1.2e8



The explanation for these oscillatory perturbations is orbital resonance. The gravitational effect of Mimas on the particle in the Cassini division is periodic. So, every revolution around Saturn, when the moon and the particle align, the particle is pulled slightly more by the moon than by Saturn. It is not long enough to pull it completely from its orbit; however, if repeated a few times, it eventually gets thrown off(space wiki). It can be observed on the Figure.4 by the "humps" when the moon pulls the trajectories. Then, they go back to 0 separation as they move out of the moon's effect back to orbiting Saturn and then back on the peak again. After a few iterations, the particles eventually fly out of the orbit, and the displaced trajectories diverge.

Finally, we can calculate the Lyapunov exponent for all distances from Saturn between 0.6e8 and 1.4e8 metres and plot it against these distances (Figure 5).

Figure 5: Lyapunov exponent simulation results vs Saturn's rings picture



Even though the simulation and the picture do not align perfectly, there are a lot of similarities and common behaviours. Following the definition of the Lyapunov exponent, the higher it is, the less stable the chaos. In our graph, it would translate: the higher the peak, the more unstable the orbit, so the darker the gap in the ring and vice versa. If we look closely enough, the largest and most important gaps align almost perfectly with the real-life observations with an accuracy of about $\pm 0.05e8$

Figure 5. accurately represents:
- The rings starting at about 0.75e8 ending at 1.34e8
- The position of ring A - starts at a distance of about 1.22e8 and ends at 1.34e8
- Position of Cassini Division - for distances 1.17e8 - 1.2e8 no stable troughs
- Position of all major gaps - Maxwell gap at about 0.82e8, Cassini division, Encke gap at about 1.31e8, and a big gap from 1.36e8 - 1.4e8.
- Position of most of the small gaps in B ring - distances 0.9e8 - 1.17e8

The main differences and inaccuracies:
- Too many unstable regions for A ring
- Pretty much chaotic behaviour for values up to the first major gap - Maxwell gap.
- Lyapunov exponent peaks grow linearly, showing there is some systematic error in the simulation.
- All points are chaotic, having a positive Lyapunov exponent. They all diverge in less than 10 million seconds (115 days), contrary to Saturn's rings, which are stable for hundreds of millions of years.

4 Evaluation

There are a few explanations for the inaccuracies and errors. First, the whole system of rings was reduced to Saturn, a particle of negligible mass, and the closest moon, Mimas. In reality, Saturn has many moons that influence the shape of its rings, and the rings themselves have a lot of mass that influences motion. If more moons were taken into account, the behaviour for up to 0.8 would probably be more stable as more force would pull on it from a distance. Moreover, the rings having non-negligible mass could solve the problem of diverging in 100 days and the stable rings not having many stable regions. Finally, for all positions r from Saturn, the initial conditions had the initial velocity $vy = 2e4$ m/s. If this value varied with distance based on the orbital velocity formulae, it would fix the issue of Figure.5 being mostly accurate for the B-ring. However, it is not easily calculated since two bodies (Saturn and Mimas) are pulling on it from two different sides.

5 Conclusion

Nonetheless, all the simulations and graphs produced show that the behaviour and shape of the rings of Saturn are chaotic. All graphs show accurately the positions of the most important characteristic points on the rings. They thus show that the rings have to look like they do since the particles placed in the gaps will immediately diverge due to the resonance with Mimas. Particles placed on stable rings will take much longer to diverge. Considering that in real life, the rings have a big mass, in the stable regions, the particles will be held by their ring's gravity and not diverge as quickly as in simulations.

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